

Multivariate Adaptive Regression Splines (MARS)

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General Outline of the MARS Method

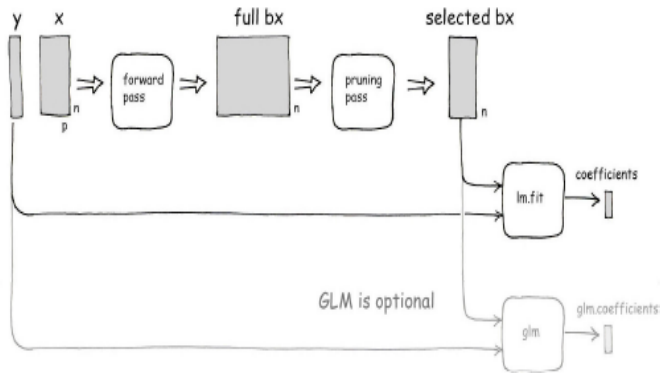
- ❶ The MARS (Multivariate Adaptive Regression Splines) method was proposed by J. H. Friedman.
- ❷ A highly flexible method of regression
- ❸ The greatest advantage of the MARS method is its ability to identify transformations of the original variables and interactions between variables.
- ❹ A non-commercial implementation of the method is available in the following packages: **earth** in R and **py-earth** in Python.

General Outline of the MARS Method

The procedure consists of three steps:

- ➊ The **FORWARD** step involves the iterative creation of new variables, based on variables generated in previous iterations. In each step, two new variables are added. The procedure is stopped if one of the stopping conditions is met. As a result of the FORWARD step, a model is obtained that typically contains too many variables and is overfitted to the training data. Therefore, variable elimination is necessary.
- ➋ The **BACKWARD** step involves stepwise elimination of variables found during the FORWARD step. In this step, we identify a model (i.e., a subset of variables) that minimizes a certain criterion function (e.g., GCV – Generalized Cross-Validation).
- ➌ **MODEL FITTING** based on the subset of variables identified in Step 2. In this step, we can use any regression or classification model.

General Outline of the MARS Method



Scheme of MARS method (source: R package `earth`).

Forward step

- Let us consider iteration number $l = 1, 2, \dots$
- Let $B = [B_1, \dots, B_M]$ be an $n \times M$ matrix whose columns contain the variables $B_1, \dots, B_M$ determined in the previous iterations $1, \dots, l - 1$.
- The functions $B_1, \dots, B_M$ are referred to as the 'parents'.
- We have that $M = 2l - 1$.
- The first column of the matrix B is always constant and equal to 1, i.e., $B_1 = 1$.

Forward step

- The method uses hinge functions $(t - x_v)_+$, $(x_v - t)_+$.
- In iteration l **we find two new variables ("children")**, which take the form:

$$B_l \cdot (t - x_v)_+$$

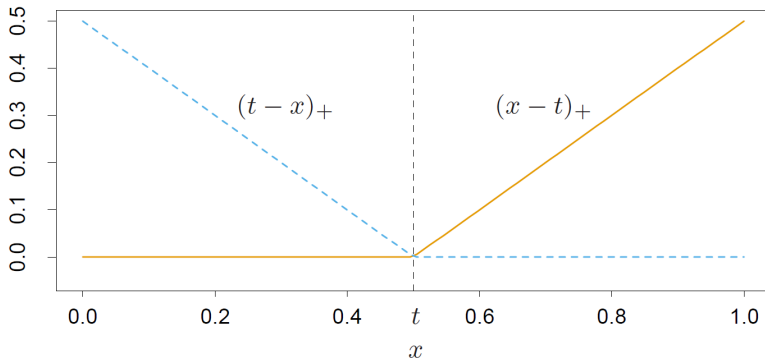
and

$$B_l \cdot (x_v - t)_+,$$

where:

- $l \in \{1, \dots, M\}$ is the selected parent,
 - x_v is the selected explanatory variable,
 - t is the selected threshold for the variable x_v
- Note that for $t = 0$, the function $(x_v - 0)_+ = \max(0, x_v)$ (RELU).

Forward step



Hinge functions used in MARS (source: "The elements of statistical learning").

Forward step

In each iteration we have to choose the optimal values of l, v, t .

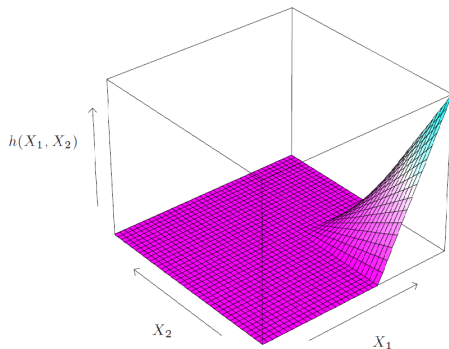
We minimize the following objective function:

$$(l^*, v^*, t^*) = \arg \min_{l, v, t, a_1, \dots, a_{M+2}} RSS(l, v, t, a_1, \dots, a_{M+2}),$$

gdzie:

$$RSS(l, v, t, a_1, \dots, a_{M+2}) = \sum_{i=1}^n \left[y_i - \sum_{j=1}^M a_j B_j - a_{M+1} B_l \cdot (t - x_v)_+ - a_{M+2} B_l \cdot (x_v - t)_+ \right]^2.$$

Interactions in MARS



Interactions of the form: $(X_1 - t_1)_+ \cdot (X_2 - t_2)_+$ (source: "The elements of statistical learning").

Backward step

Suppose that in the FORWARD step we have determined the variables $B_1, \dots, B_{M_{\max}}$. We want to find a subset $m \subset \{B_1, \dots, B_{M_{\max}}\}$ that minimizes the generalized cross-validation criterion:

$$GCV(m) := \sum_{i=1}^n \left[\frac{y_i - \hat{y}_i(m)}{1 - \frac{C(m)}{n}} \right]^2,$$

where:

- $\hat{y}_i(m)$ is the prediction for the i -th observation, computed based on model m
- $|m|$ is the number of (non-constant) variables in model m
- $C(m) = 1 + |m| + \text{pen} \cdot |m|$ (where pen is an additional penalty parameter for the number of variables in model m).

Backward step

Remarks:

- Computing GCV for all subsets m of the set $\{1, \dots, M_{\max}\}$ is computationally expensive.
- Therefore, we use the standard stepwise variable elimination procedure. In the first step, we fit the model using all variables. In the next step, we remove the variable whose omission causes the largest decrease in GCV . The procedure is continued until removing any variable no longer improves the model (i.e., does not decrease GCV).
- Alternatively, other criterion functions can be used (e.g., AIC/BIC) and other subset searching methods (e.g., forward selection).

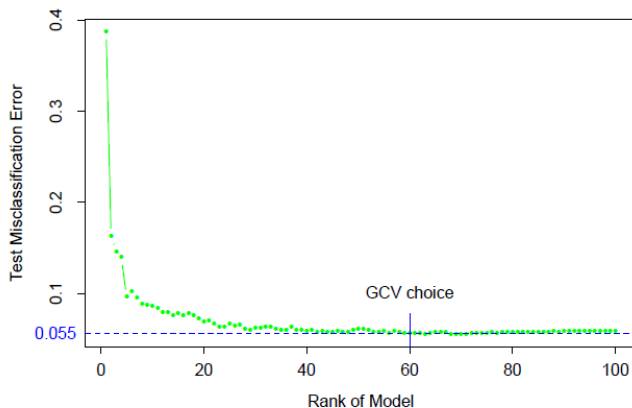
MODEL FITTING

- Based on the variables $B_{i_1}, \dots, B_{i_K}$ created in the FORWARD step and selected in the BACKWARD step, we can fit the final model

$$Y \sim B_{i_1}, \dots, B_{i_K}$$

- In the R package, a linear regression model (using the `lm` function in R) or a logistic regression model (using the `glm` function in R) is used, depending on whether the variable Y is quantitative or binary.
- Any regression or classification method can be used in this step (e.g., decision trees, ensembles, neural networks, etc.).

MARS: example on Spam dataset



Classification error with respect to the number of variables $B_1, B_2, \dots, B_{100}$ (źródło: 'The Elements of Statistical Learning').

Conclusions

- The MARS method can be used as a tool for creating a new representation of variables.
- The MARS method allows for the creation of variable transformations and the identification of interactions between variables (including interactions of order greater than 2).
- In the final third step of the procedure, any classification/regression model can be used.
- The method allows modeling of nonlinear relationships between the response variable and the explanatory variables.